

class Student:
def __init__(self, name, sap_id, phy, chem, maths):

self.name = name

self.sap_id = sap_id

self.phy = phy

self.chem = chem

self.maths = maths

def display(self):
 print(f"Name: {self.name} sap Id:
 {self.sap_id}")
 print(f"Marks → Physics:
 {self.phy}, Chemistry: {self.chem}, Maths
 {self.maths}")
 print()

students = []

for i in range(3):

name = input("Enter name:")

sap = input("Enter sap Id:")

phy = int(input("Physics marks:"))

chem = int(input("Maths Marks:"))

students.append(Student(name, sap, phy, chem, maths))

for s in students:

s.display()

Output

Enter name = Poigamohu

Enter SAP ID = 590025830

Physics marks: 75

Chemistry marks: 88

Maths: 98

Enter name: test

Enter SAP ID: 590025830

Physics marks: 85

Chemistry marks: 45

Maths marks: 98

Name: Shagan, SAP ID: 590025830

Marks → Physics: 78, Chemistry: 85, Maths: 50

Name: test, SAP ID: 590025830

Marks → Physics: 85, Chemistry: 75, Maths: 50

Name: test 2, SAP ID: 590025830

Marks → Physics: 65, Chem: 50, Maths: 98

② Class Student:

```
def __init__(self, name, sap, p, c, m)
```

```
    self.name = name
```

```
    self.sap = sap
```

```
    self.marks = [P, C, M]
```

```
def __display(self)
```

```
    print(f"Name {self.name} SAP {self.sap}")
```

```
    print("Marks: ", self.marks)
```



```

def percentage (self)
    return sum (self.marks) / len(self.marks)

def result (self)
    return 'Pass' if all (m >= 50 for m in self.marks) else 'Fail'

students = []
n = int (input ("No. of student: "))
for i in range (n)
    print (f"Enter student {i+1} ")
    name = input ("Name: ")
    sap = input ("SAP: ")
    f, c, m = map (int, input ("Enter 3 marks: ").split ())
    student.append (student (name, sap, f, c, m))

for s in students
    s.display ()
    print (f"1: {s.marks} percentage: {s.percentage ()} 2: {s.result ()}")
print ("In class avg: sum(s.percentage) / len(students) / n")

```

Output

No. of student = 2

Student 1

Name: Poigamohu

SAP: 580025030

Enter 3 marks: 75, 59, 63

Student 2

Name : Jett

SAP ID : 590025230

Enter 3 marks : 75, 60, 88

Polygonshu (590025230)

Marks : 75 60, 50

% : 80.33% Pass

Test (590025230)

Marks : (60 68 70)

% : 88% Pass

Class avg : 80.6000

① Class parents

```
def show(self):
```

```
    print("Parent class")
```

```
class child (Parent)
```

```
    def display (self)
```

```
        print("Parents class")
```

```
class child (Parents)
```

```
    def display (self)
```

```
        print("Child class")
```

```
obj - child()
```

```
obj.show()
```

```
obj.display()
```

```
def show X(self)
```

```
    print("Class X")
```

Class B:

```
def show B (self)
    print ("class B")
class C:
    def show (self):
        print ("class C")
```

```
obj = C()
obj.show A()
obj.show B()
obj.show C()
```

Output

Parent class

child class

class A

class B

class C

class A

class B

class C

⑤ class Vehicle:

```
def info (self):
    print ("This is a vehicle")
```

```
class Bike (Vehicle):
```

```
    def info (self):
        print ("This is bike")
```

```
V = Vehicle()
```

```
b = Bike()
```



```

print('Vehicle object is')
info()
@ print('Bike object (overridden method):')
info()

```

Output

Vehicle object

This is a vehicle

Bike object (overridden method):

This is a bike

⑤ class Point:

```

def __init__(self, x, y):

```

```

    self.x = x

```

```

    self.y = y

```

```

    def __add__(self, other):

```

overloading

```

        print("Adding two point objects")

```

```

        return Point(self.x + other.x, self.y + other.y)

```

```

    def display(self):

```

```

        print(f"({self.x}, {self.y})")

```

```

p1 = Point(10, 20)

```

```

p2 = Point(12, 10)

```

```

print("P1:", end=" ")

```

```

p1.display()

```

```

print("P2:", end=" ")

```

```

p2.display()

```

```

p3 = p1 + p2

```

```

print("P3 (P1 + P2):", end=" ")

```

```

p3.display()

```

Output

$P_1 : (10, 20)$

$P_2 : (12, 15)$

Adding two point object

$P_3 (P_1 + P_2) : (22, 35)$